# HDOC Day-73

Enumeration (enum) in C

## 1. Enumeration for Days

Create an enumeration for days (SUNDAY to SATURDAY) and print each day with its integer value.
**Input:** No Input
**Output:** SUNDAY = 0, MONDAY = 1, TUESDAY = 2, WEDNESDAY = 3, THURSDAY = 4, FRIDAY = 5, SATURDAY = 6

### C Program

```
#include <stdio.h>

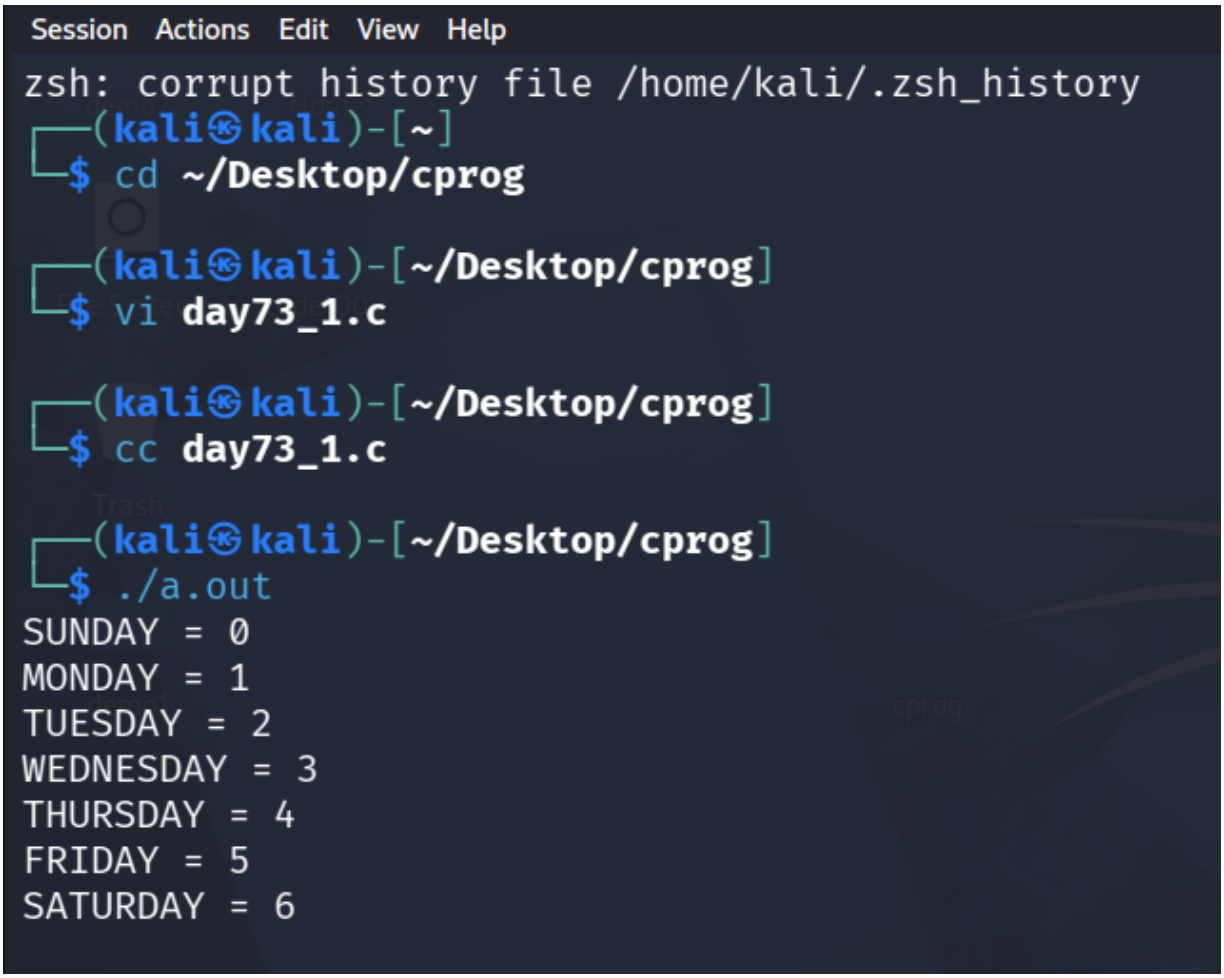
enum Days {
    SUNDAY,
    MONDAY,
    TUESDAY,
    WEDNESDAY,
    THURSDAY,
    FRIDAY,
    SATURDAY
};

int main()
{
    printf("SUNDAY = %d\n", SUNDAY);
    printf("MONDAY = %d\n", MONDAY);
    printf("TUESDAY = %d\n", TUESDAY);
    printf("WEDNESDAY = %d\n", WEDNESDAY);
    printf("THURSDAY = %d\n", THURSDAY);
    printf("FRIDAY = %d\n", FRIDAY);
    printf("SATURDAY = %d\n", SATURDAY);
    return 0;
}
```

### Test Case

| Test Case | Input | Expected Output | Result |
|---|---|---|---|
| 1 | No Input | SUNDAY = 0 through SATURDAY = 6 | Pass |

### Kali Terminal Run


Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~]
$ cd ~/Desktop/cprog
(kali㉿kali)-[~/Desktop/cprog]
$ vi day73_1.c
(kali㉿kali)-[~/Desktop/cprog]
$ cc day73_1.c
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
SUNDAY = 0
MONDAY = 1
TUESDAY = 2
WEDNESDAY = 3
THURSDAY = 4
FRIDAY = 5
SATURDAY = 6

## 2. Enumeration for Traffic Lights

Define an enumeration for traffic lights (RED, YELLOW, GREEN) and print 'Stop', 'Wait', or 'Go' based on its value.

**Input:** GREEN

**Output:** Go

### C Program

```
#include <stdio.h>

enum TrafficLight {
    RED,
    YELLOW,
    GREEN
};

int main()
{
    enum TrafficLight light = GREEN;

    printf("RED = %d\n", RED);
    printf("YELLOW = %d\n", YELLOW);
    printf("GREEN = %d\n", GREEN);

    if (light == RED)
        printf("Stop\n");
    else if (light == YELLOW)
        printf("Wait\n");
    else if (light == GREEN)
        printf("Go\n");

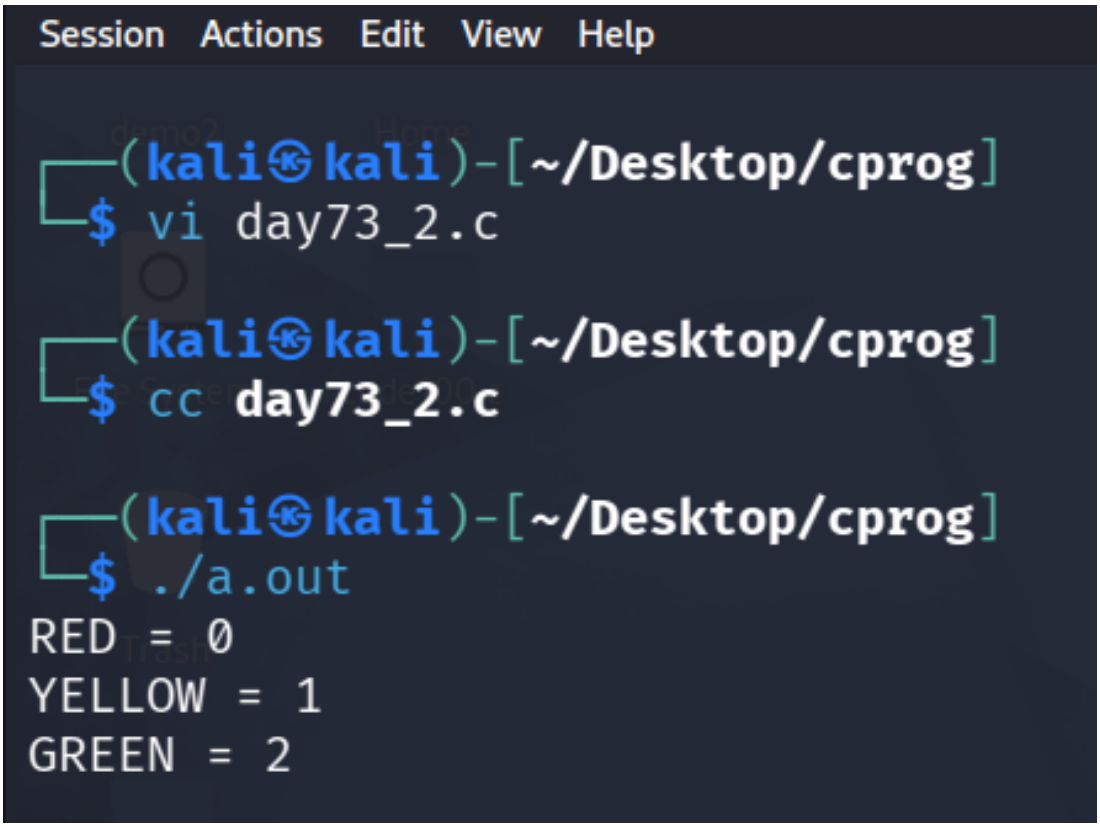
    return 0;
}
```

### Test Cases

| Test Case | Light | Expected Message |
|---|---|---|
| 1 | RED | Stop |
| 2 | YELLOW | Wait |
| 3 | GREEN | Go |

### Provided Kali Terminal Run

The supplied terminal screenshot shows the enum integer values: RED = 0, YELLOW = 1, GREEN = 2.


Session Actions Edit View Help
(kali㉿kali)-[~/Desktop/cprog]
$ vi day73_2.c
(kali㉿kali)-[~/Desktop/cprog]
$ cc day73_2.c
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
RED = 0
YELLOW = 1
GREEN = 2